



Maritime &
Coastguard
Agency

Guidance

MGN 379 (M+F) Amendment 1: use of electronic navigational aids

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Summary

This Notice emphasises the need for correct use of electronic navigational equipment by watchkeepers.

Key Points:

- Be aware that each item of equipment is only a navigational aid on board.
- Be aware of the factors which can affect the accuracy of electronic position fixing systems.
- Appreciate the need to cross check and verify position fixing information using other methods, including non-electronic.
- Recognise the importance of the correct use of navigational aids and knowledge of their limitations.
- Be aware of the dangers of over-reliance on the output from, and accuracy of, a single navigational aid.
- This amendment contains updates to references, procedures, technology, and terminology which have all developed since the first iteration of this notice.

1. Introduction/background

Accidents have occurred where the primary cause has been found to be over-reliance on a single electronic navigational aid, or over-reliance on electronic aids in general, in lieu of the maintenance of a proper and effective visual lookout. Watchkeepers must always ensure that positional information is regularly cross checked and verified using other equipment, as well as visual aids to navigation.

In other cases, accidents have occurred where the watchkeeper was found not to be fully familiar with the operation of equipment or its limitations.

2. Provision of navigational equipment on ships

The Merchant Shipping (Safety of Navigation) Regulations 2020 (SI 2020/0673) implement the carriage requirements for navigational equipment set out in Regulation 19 of the International Convention for the Safety of Life at Sea (SOLAS), 1974 Chapter V, a new version of which came in force on 1 July 2002.

Ships built before 1 July 2002 may continue to comply with the requirements of SOLAS Chapter V/74 in force prior to 2002 Regulations, regarding Signalling Lamps (Reg. V/11/74), Navigation Equipment (Reg. V/12/74) and Nautical Publications (Reg. V/20/74). However, they must carry a Global Navigation Satellite System (GNSS) receiver or a terrestrial radio-navigation receiver, Automatic Identification System (AIS), Voyage Data Recorder (VDR) or Simplified Voyage Data Recorder (S-VDR), and a bridge navigational watch alarm system (BNWAS). In addition, when on international voyages they may also be required to carry an electronic chart display and information system (ECDIS).

3. RADAR

3.1 General

Radar is an important tool for the watchkeeper in all conditions of visibility. Despite the principles remaining broadly the same since its inception,

modern iterations of radar equipment include new features and technologies, and the mariner should be fully aware of their function and associated limitations. Incorrect maintenance, set up, and operating procedures can severely limit the effectiveness of radar equipment hence the importance of equipment familiarisation and cross-checking of inputs to and outputs from the radar.

Collisions have frequently occurred due to failures in making effective use of radar and radar plotting aids in both restricted visibility and good visibility. Common errors noted are, altering course based on insufficient information, and maintaining speed in excess of the safe speed that may be required, particularly when a close-quarters situation is developing. Information provided by radar and radar plotting aids in clear weather conditions can assist the watchkeeper in support to maintaining a proper lookout in areas of high traffic density. It is most important to remember that navigation in restricted visibility can be more demanding and great care is needed due to copious information available from various electronic aids to navigation, including radar and associated automatic target tracking aids. Where continuous radar watchkeeping and plotting, or target tracking, cannot be maintained an even greater caution must be exercised. A “safe speed” should be maintained at all times with explicit reference to rule 6 of the International Regulations for Preventing Collisions at Sea 1972 (hereby referred to as COLREG), for vessels operating with radar.

3.2 Electronic radar plotting and tracking aids

Radars are equipped with plotting aids, the type of which depends upon the size of ship as follows:

(i) Electronic Plotting Aid (EPA) EPA equipment enables electronic plotting of at least 20 targets, but without automatic tracking (Ships of 300 Gross Tonnage (GT) and over and less than 500 GT).

(ii) Automatic Tracking Aid (ATA) ATA equipment enables manual acquisition and automatic tracking and display of at least 30 targets (Ships of 500 GT and over). On ships of 3,000 GT and over, the second radar must also be equipped with an ATA. The two ATAs must be functionally independent of each other.

(iii) Automatic Radar Plotting Aid (ARPA) ARPA equipment, required on ships of 10,000 GT and over, provides for manual or automatic acquisition of targets and the automatic tracking and display of all relevant target information for at least 40 targets for collision avoidance decision making.

It also enables trial manoeuvres to be executed. The second radar must incorporate ATA if not ARPA.

Manual radar plotting equipment is no longer acceptable except for existing vessels still complying with SOLAS V/74.

Watchkeepers must be fully familiar with the operation and limitations of these automatic target tracking and plotting facilities and should practice using them in clear weather conditions to improve their skills.

Due regard should be given to the time required for an automatic plotting system (such as ARPA) to acquire a target, and then accurately calculate data and solutions. This can be up to 3 minutes before full and smoothed accuracy is reached, and action before this timeframe may carry higher risk. Use of radar does not reduce the requirement of a watchkeeper to be identifying objects and hazards early. These minimum tolerances of error should also be known, especially for CPA, as when approaching at close quarters the CPA may be near zero despite a greater indicated value by the radar.

In addition to the advice given above and the instructions contained in the operating manual(s), users of radar plotting aids should ensure that:

- (i) performance of the radar is monitored and optimised; and
- (ii) test programmes, if provided, are used to check the validity of the plotting data; and
- (iii) speed and heading inputs to the ARPA/ATA are satisfactory. Correct speed input, where provided by manual setting of the appropriate ARPA/ATA controls or by an external input, is vital for correct processing of ARPA/ATA data. Serious errors in output data can arise if heading and/or speed inputs to the ARPA/ATA are incorrect.

For full guidance and regulation on carriage requirements see **MGN 610** and SOLAS Chapter V/19

3.3 Monitoring of targets

Radar should be used to assist in the assessment of whether risk of collision exists, or a close quarters situation is likely to develop, and in clear weather it can complement visual observations.

To estimate risk of collision with another vessel, the closest point of approach (CPA) and Time to Closest Point of Approach (TCPA) should be established in ample time. Choice of appropriate avoiding action is facilitated by the knowledge of the other vessel's course using the manual or automatic plotting methods. The accuracy of the plot, however obtained, depends upon accurate measurement of own ship's heading and speed, which is inputted into the radar equipment by suitable external sensors during the plotting interval. It is important to note that inaccurate compass heading or speed input will reduce the accuracy of target vessel's true vectors when using ARPA or ATA. This is particularly important with targets on near reciprocal courses where a slight error in own ship's data input may lead to a dangerous interpretation of the target vessel's true course. The perceived precision of digital readouts should, therefore, be treated with caution.

If two radars are fitted, it is good practice, especially in restricted visibility, congested, or shallow waters, for one to be designated for anti-collision work, while the other is used to assist with navigation. If only one of the radars is fitted with automatic target tracking, then this should be the one used for anti-collision work and the other for navigation.

A single observation of the range and bearing of an echo will give no indication of the relative course or heading of the other vessel. To estimate this, a succession of systematic observations must be made over a known time interval. The longer the period of observation, the more accurate the result will be. This also applies to ARPA/ATA which require adequate time to produce accurate information suitable for assessing CPA/TCPA and determining appropriate action.

Estimation of the target's true course is only valid up to the time of the last observation and the situation must be kept constantly under review. The other vessel, which may not be keeping a radar watch or plotting, may subsequently alter its course and/or speed. This will take time to become apparent to the observer. Electronic plotting will not detect any alteration of a target's course or speed instantly and, therefore, should be checked by other means.

The compass bearing, observed either visually or by radar, should be used to assess risk of collision. The relative bearing of a target should not be used when own ship's course and/or speed alters, as risk of collision may still exist even where the relative bearing is changing. Mariners should also be aware that at close range, risk of collision may still exist even with a changing compass bearing.

Where AIS target information is available as an overlay on the radar in use, care should be taken when selecting a target to ensure that the target information shown is the ARPA produced data rather than AIS, due to

potential inaccuracies in AIS data as well as AIS utilising course/speed over ground rather than through the water, which can produce an inaccurate aspect of the target to the observer. AIS data may be used for aiding situational awareness, but not relied upon for collision avoidance. Further information is detailed in section 6 of this guidance note.

3.4 Verifying equipment performance.

It is essential for the operator to be aware of the radar's current performance, which can be ascertained following the guidance given in the IMO performance standards, including the use of a Performance Monitor. Masters and navigating officers should refer to guidance given by the equipment manufacturer for details regarding performance checks and their frequency including the use of the Performance Monitor, if fitted. It is recommended that maintenance routines are recorded within the vessel's planned maintenance system and/or scheduled procedures are clearly defined within the safety management system.

The observer should be aware of the arcs of blind and shadow sectors on the display caused by masts and other on-board obstructions. It is recommended that these zones be plotted on a diagram and placed near the radar display. This diagram should be updated following any changes which affect the sectors.

When radar is used to complement visual observations in clear weather, it allows radar observations and the resulting electronic vectors to be checked. It will assist in determining any misinterpretation of the radar display or a misleading appraisal of the situation, which could potentially be dangerous in restricted visibility, or in other situations where radar is relied upon more heavily.

3.4.1 Operational checks

Regular checks of the radar settings, detection and the picture set up should be made to ensure that the quality of the display and the performance of the radar has not deteriorated. The mariner should be familiar with the specifics of which other equipment provides the data and input feed to the radar, for example speed log and gyro.

Misalignment of the heading marker, even if only slight, can lead to a misleading interpretation of potential collision situations, particularly in restricted visibility when targets are approaching from ahead or fine on the bow. It is therefore important that checks of the heading marker should be made periodically to ensure that correct alignment is maintained. If misalignment exists, it should be corrected at the earliest opportunity. The

manufacturer's user manual should be referred to for equipment specific methods of carrying out heading marker checks.

3.5 Choice of settings and parameters

It is important that the mariner understands the parameters of the radar set(s) fitted to their vessel and, in the case where there are multiple sets, the key differences between them. Many ships are fitted with both X and S band radars and good bridge management should ensure that the differences, as well as the pros and cons of each type of radar are known to the user.

X-Band Radar	S-Band Radar
3cm – 9GHZ	10cm – 3GHZ
Ability to detect a 9GHZ SART	Improved target detection in heavy weather
Improved small target detection	Improved longer range detection
Higher resolution image due to high frequency, improving coastline identification during navigation	Improved sea clutter response

The choice of range scale is important on any radar set, and periodic scanning at a longer-range scale will allow advance warning of hazards. Acquiring targets on a short-range scale may not allow enough time to appreciate the risk of collision and take appropriate action as necessary. This applies particularly when approaching areas where high traffic density is likely or when an early appraisal obtained from the use of longer-range scales may be an important factor in determining safe speed. More generally, the choice of range scales for observation and plotting is dependent upon several factors such as traffic density, speed of own ship, prevailing weather conditions, proximity of navigational hazards, and the frequency of observation.

Modern radar sets may include automatic selection of pulse length based on the range scale in use. However, this should be checked by the operator, and for radars without this feature the watchkeeper should ensure that the pulse length in use is appropriate to the range scale. In

general, this will be directly proportionate with short pulse lengths used for shorter range scanning and vice versa. The watchkeeper should be aware of how longer pulse lengths, whilst allowing greater detection at range, may obscure targets in close proximity to each other due to the reduced resolution of the beam.

Echoes may be obscured by sea or rain clutter. Correct setting of clutter controls will help but may not completely remove this possibility. Sea, rain, and gain settings should be properly adjusted on setup and constantly monitored, especially when changing range scales.

3.6 Stabilisation and Motion

The watchkeeper should be aware of both the stabilisation mode and motion mode being utilised by each radar set in use and the differences between each mode.

3.6.1 Stabilisation

The navigator should determine which stabilisation mode is the most suitable for the intended use.

Ground Stabilisation	Sea Stabilisation
Speed and Heading from GNSS	Speed and Heading from Gyro (or another THD) and Log
Course and speed over ground	Course and speed through the water
Best suited for coastal Navigation	Best suited for Anti-collision
Set and Drift visually shown	Does not account for set and drift
Difficulty in determining true aspect of vessel	True aspect of vessel easily determined
A fixed target will appear stationary	Fixed target will have a vector direction reciprocal to set and the speed will indicate the rate of drift

In basic terms, a radar that is sea-stabilised uses the speed input from the waterlog, giving speed through the water (STW); and a radar that is ground-stabilised takes speed input from the GNSS, giving speed over the ground (SOG).

UK guidance is that sea-stabilised is the preferred mode for open sea and anti-collision decision making, this is because set and drift will not be taken into account within the radar ARPA calculation, and therefore the resulting true vector of a target will provide a more accurate representation of other vessels' aspect.

However, fixed objects such as a landmass, buoys, or anchored vessels, when in sea stabilised mode, will show trails in true motion, this makes it difficult to differentiate between fixed and stationary targets and may cause excessive clutter and potential confusion to the operator. Setting the radar to ground stabilised will avert this and allow the watchkeeper greater differentiation between fixed or stationary objects, and those which are making way. Ground stabilised mode will also give the watchkeeper a clear visual indication via a true vector originating from their vessel, of their own set and drift. This is likely to improve situational awareness during coastal and pilotage navigation, though the operator should be aware that the application of set and drift at their location will also be applied to targets, which may mean the expected aspect of a target as seen on the radar will be different to a visual determination.

3.6.2 Motion (Relative/True)

The watchkeeper should be aware of the motion mode selected and have a full appreciation of the differences between both.

When true motion is used, the displayed position of the navigator's own ship moves across the radar display at a speed corresponding to the vessels actual motion. In relative motion mode however, the displayed position of the navigator's own ship will remain static.

3.6.3 Use of Target Trails

The navigator should be aware of the different target trail options and have an appreciation of the difference in the visual indications given by each mode.

With relative trails, the relative movement of other vessels based on the combined movement of both vessels is displayed. With True trails however, the trails show the true movement of targets depending on their course and speed over the ground. A fixed object such as land or fixed navigational mark will not produce a true trail.

Relative trails can give a quick indication to the navigator with regards to a risk of collision but can also introduce unnecessary trails from fixed targets

such a landmasses or anchored vessels.

3.6.4 Display orientation modes

When setting up the radar display, the operator should be aware of the different display modes available and the differences between them.

In head-up (HU) mode, the 'up' direction of the display represents the vessels heading but is 'un-stabilised' as it does not utilise a heading input; in course-up (CU) it represents the direction that has been selected as the vessels course; and in North up (NU), the display is orientated so that the top of the screen is north, similar to a standard ECDIS display.

North-up display is often preferred in coastal navigation due to the representation matching an ECDIS or chart display. During collision avoidance however, a course-up display may be preferred due to the display matching what the navigator can see visually.

North up and Course Up display modes require a heading input from a Gyro or transmitting heading device (THD). If a failure of the heading input is experienced, the radar will revert to a head-up, un stabilised mode which could cause significant smearing of land and other echoes with alterations of course.

3.7 Operation

Radar, if fitted, should be operating at all times, regardless of traffic density or state of visibility, unless there is an overriding safety or operational reason. Radars are designed for continuous operation and frequently switching them on and off may cause technical difficulties, as well as erasing tuning settings. When weather conditions indicate that visibility may deteriorate, and at night when small craft or unlit obstructions such as a derelict hull or ice are likely to be encountered, both radars, if fitted, should be operating, with one dedicated to collision avoidance work. This is particularly important when there is a likelihood of occasional fog banks, so that vessels can be detected before entering fog. Careful consideration should be given to the required application and use of radar as referenced in COLREG, especially when in restricted visibility and/or making assessment of a risk of collision.

Use of automatic detection/tracking features may assist in reducing workload but should not be wholly relied upon and careful checking and correlation with visual lookout should be maintained. Whether using automatic features or not, the watchkeeper should consider regular clearing down of selected targets that have passed and are no longer

considered a risk. This will avoid confusion, cluttered displays, unnecessary alarms, and possible technical malfunction.

Some radars are provided with electronic chart overlays. These charts may have a limited amount of data and are not the equivalent to an official Electronic Navigational Chart (ENC) used in the ECDIS, or paper charts; therefore, they should not be used as the primary basis for navigation.

These should not be confused with Radar Image Overlay (RIO) which can be used on ECDIS to verify GNSS quality by aligning the radar image with an official ENC.

When using radar for position fixing and monitoring, the following should be considered and checked:

- (i) the identity of fixed objects,
- (ii) tidal variations and ice can significantly alter the outline of the coast and therefore the accuracy of position fixing and monitoring,
- (iii) the radar's overall performance,
- (iv) the gyro error and accuracy of the heading marker alignment,
- (v) that parallel index lines are correctly positioned on a suitable display; and
- (vi) the accuracy of the variable range marker, bearing cursor and fixed range rings.

3.8 Parallel Indexing

Parallel Index techniques provide the means of continuously monitoring a vessel's position in relation to a pre-determined passage plan, and therefore a way to either monitor GNSS integrity, or navigate in the absence of GNSS. Parallel indexing should be practised in clear weather during straightforward coastal passages, so that watchkeepers remain thoroughly familiar with the technique and confident in its use in more demanding situations, for example in confined waters, restricted visibility, darkness, or in situations of GNSS denial.

The principles of parallel index plotting can be applied, using electronic index lines. A number of index lines may be pre-set and called up when required, but features will vary between different types and models of

radar set. Care should be exercised when activating pre-set parallel index lines to ensure that the correct line(s) for the passage are being displayed.

3.8.1 Parallel indexing technique

Parallel Indexing (P.I.) can be used on both Sea and Ground-stabilised radar displays. Reference is first made to the chart and the planned track. The index line is drawn parallel to the planned track at a perpendicular distance (CIR or offset) equal to the planned passing distance off an appropriate fixed target. Observation of the fixed object's echo moving along the index line will indicate whether the ship is maintaining the planned track. Any displacement of the echo from the index line will immediately indicate that own ship is not maintaining the desired track, enabling corrective action to be taken.

Parallel Indexing may be used in either Relative or True Motion, however it should be noted that the P.I Line needs to be referenced to the mariner's own vessel. In relative motion this can be the same as being fixed to the screen centre, but for True Motion it would need to move with the vessel and hence cannot be fixed to the screen.

3.8.2 Integration with ECDIS

Where the radar display is integrated with an Electronic Chart Display and Information System (ECDIS), the practice of parallel indexing continues to enable the navigator to monitor the ship's position relative to the planned track on the radar, and additionally provides a means of continuously monitoring the positional integrity of the ECDIS system.

3.8.3 Precautions

Some older radars may still have reflection plotters. It is important to remember that parallel index lines drawn on reflection plotters apply to one range scale only. In addition to all other precautions necessary for the safe use of radar information, particular care must therefore be taken when changing range scales.

The use of parallel indexing does not remove the requirement of position fixing at regular intervals using all appropriate methods available including visual bearings. Parallel indexing only indicates if the ship is on or off track and not its progress along the track.

3.9 Speed Log Failure

To determine a target's aspect by radar and to complete the calculation of its true heading or course, either automatically or manually, is dependent

on the choice and accuracy of the own ship's speed input. On the assumption that both a Speed log and GNSS receiver are connected to the radar, failure of a speed log will result in the radar only being able to determine Speed over Ground. The mariner should therefore be familiar with how to identify such a failure mode and understand the difference between sea and ground stabilised modes as described in Section 3.6 when determining if a risk of collision exists.

3.10 Gyro failure

In cases of gyrocompass failure and when the radar's heading data is provided from a transmitting magnetic compass (TMC), watchkeepers should determine and apply the magnetic compass error(s).

The true vector function of automatic plotting and tracking equipment should be operated with caution when the heading input is derived from a Transmitting Magnetic Compass (TMC). Target tracking prediction is reliant on steady state tracking, based on the assumption of course and speed remaining steady. In a seaway, a transmitting magnetic compass may not produce a sufficiently steady heading resulting in unreliable vectors. A failure of all heading inputs into the radar will result in the display mode returning to an un stabilised head up mode (See section 3.6.4)

3.11 Warnings and alarms

Audible operational warnings and alarms may be used to indicate that a target has closed on a pre-set range, enters a user-selected guard zone, or violates a pre-set CPA or TCPA limit. When the ARPA is in automatic acquisition mode, these alarms should be used with caution, especially in the vicinity of small radar-inconspicuous targets. Users should familiarise themselves with the effects of error sources on the automatic tracking of targets by reference to the ARPA operating manual. Such alarms do not relieve the mariner of their duty to maintain a proper lookout by all available means, including sight and hearing.

3.12 Evolving Technologies

There are increasing numbers of new and evolving radar technologies entering the market, especially in the sector for vessels not requiring full SOLAS compliance. Prevalent is the use of solid-state radars, and methodology such as pulse compression. Watchkeepers may also see far more assistance features built into ARPA or radar software, but in any case, the principles of radar navigation remain unchanged and the watchkeeper should continue to track and plot targets, cross check them visually or by other means, and utilise all tuning and range features to ensure they can maintain full appraisal of the situation and of any risk of collision.

It is also the responsibility of the Mariner and the Ship owner to ensure that adequate familiarisation, training, and procedural documentation is available when utilising a new form of Radar technology.

4. Electronic Positioning Systems

4.1 General

Ships are required to carry a Global Navigation Satellite System (GNSS) receiver or a terrestrial radionavigation system receiver. At the present time, no terrestrial radio navigation systems are operational in UK waters. Like many other electronic aids, electronic positioning systems have become cheaper and more readily available over past decades and as such are frequently used as the primary positioning source. With the increase in the number of manufacturers and satellite constellations, and subsequent reliance on the systems, there are many factors for the watchkeeper to consider when using this equipment to ensure accurate position of the vessel is not compromised.

4.2 Global Navigation Satellite Systems (GNSS)

When navigating in confined waters, navigators must bear in mind that the raw position displayed from a satellite positioning system may be that of the receiver antenna rather than centre point of the vessel. Therefore, the navigator should be aware of the actual antenna position or of any offsets inputted into either the receiver interface or external systems which utilise

the GNSS. This is especially relevant on vessels where the antenna may have a large offset from the vessel centre point.

Referred accuracy for any system relies on a clear view of the sky at the antenna and is dependent on the performance of the GNSS receiver, as well as that of the GNSS system itself. Each system has different accuracy levels (both claimed and prescribed), but it can be seen from terrestrial monitoring that GNSS usually produces horizontal accuracy within the range of 5- 25m 95% of the time. High quality receivers may see better results and augmentation systems may be applied to increase this accuracy to 1m or less. Prudent navigation should always be conducted to the limitations of the inherent accuracy and not the precision to which the receiver read out displays a position.

There are several global constellations and GNSS systems in use, developed and operated by different nations/organisations. Some provide global coverage, and some only regional. The mariner should be aware of the best coverage for the area in which they are navigating, and any associated accuracy and operational parameters that may vary from system to system; additionally, the properties of their receiver may allow for satellites from multiple systems to be used at once. Receiver equipment may require settings to be changed when changing area, and default equipment settings (based on country of manufacture) may not be suitable for the area of operation and should be checked upon initialising the system, and regularly thereafter. The mariner should also be aware that despite global coverage claims, when operating in remote areas such as the polar regions, the dispersion and angle of satellites in view might reduce accuracy and redundancy received.

(i) Global Positioning System (GPS) developed and operated by the USA and offers global coverage. Also known as NAVSTAR.

(ii) Global Navigation Satellite System (GLONASS) developed and operated by the Russian Federation and offers global coverage.

(iii) BeiDou Navigation Satellite System (BDS) developed and operated by China and offers global coverage.

(iv) GALILEO developed and operated by the European Union and offers global coverage.

(v) Indian Regional Navigation Satellite System (IRNSS) developed and operated by India and offers regional coverage across India and extending approximately 800nm from its boundaries. Global coverage is in development.

(vi)Quasi-Zenith Satellite System (QZSS)

developed and operated by Japan and offers regional coverage in the Asia-Pacific region, focussed on Japan. This system can be integrated with GPS to bolster satellite numbers and offers augmentation to the GPS system.

Volume 2 of The Admiralty List of Radio Signals, published by United Kingdom Hydrographic Office (UKHO), contains full descriptions of all GNSS systems, with notes on their correct use and limitations.

4.3 Augmentation

GNSS can be augmented to provide position error resolutions which allow greater accuracy. Ground Based Augmentation Systems (GBAS) and Satellite Based (SBAS) are the two types. Both operate with the same principle; where GNSS signals from satellites are received by a control station and checked against the station's known location. This allows errors to be calculated and corrections issued, which are then transmitted to the receiver and applied to the incoming GNSS signals to provide the final resolution. GBAS does so using a network of shore-based antenna/infrastructure whereas SBAS transmits using a network of satellites. Receivers will still work as normal when not receiving corrections but will indicate the status in some form (usually a traffic light method) and the mariner should be aware of the status and the impact this may have on positional accuracy.

Until its cessation in 2022 Differential GPS (dGPS) was the main form of GBAS in use in the UK. Many other countries used this also, but like the UK have begun to take, or have completed taking, the systems out of service. SBAS however does remain active with various systems providing regional coverage around the globe.

4.4 E-LORAN

The last hyperbolic system in use was the Long-Range Navigation system (LORAN), but this has seen rare use in recent times and has now been discontinued across the majority of the globe. Enhanced LORAN (e-LORAN) is the modern incumbent which uses advancing technology in receivers and transmitters to provide a positioning system independent of GNSS. Coverage of e-LORAN is limited globally and is not commercially available within the UK, but small-scale trials have been carried out and

there is a potential for future UK coverage. Updates to procedures and potential infrastructure in UK waters will be promulgated if and as necessary.

4.5 GNSS Reliance and checking

The prudent navigator should never rely totally on a single position fixing piece of equipment and should regularly cross check the ship's position by other, independent means. As described below, there are many factors leading to GNSS errors or uncertainty, and the watchkeeper should factor in safety margins when planning as appropriate.

Serious accidents have occurred because of over reliance on satellite positioning equipment. These include groundings and collisions caused by incorrect GNSS receiver settings, technical faults that have gone unnoticed, and the scaling up of errors which have been passed from receiver to other equipment, such as radars or auto-track systems. Over reliance is the biggest threat to the watchkeeper and, therefore, the vessel. Checking the position accuracy using other means, including visual observations, would have easily prevented these, and future, accidents.

GNSS equipment is also only as accurate as the signal it receives. In high latitudes the visibility of satellites may be compromised meaning reduced positional accuracy, or the loss of position all together. Additionally, the threat of jamming and/or spoofing of signals could mean incorrect position information is transferred onto charts/ECDIS with almost no indication of untoward behaviour to a mariner who is relying only on a single GNSS receiver. An ECDIS will only alarm when the position of the vessel fed to its system is in danger and has no way of monitoring the actual vessel position if this feed is incorrect. The only way to counter this, and remain safe, is by regular cross-checking using other available means.

4.6 GNSS Integrity and checking

GNSS signals can receive errors from many sources, including interference from other equipment, atmospheric conditions, and physical obstructions (for example signals blocked by dockside cranes, or steep terrain). GNSS position lines also work very much on the same principles

of traditional visual bearing lines, and if the cut of angle between satellites is insufficient then positional errors can be introduced.

4.6.1 Dilution of Position (DOP)

DOP is a representation of the error gained from the relative position of the satellites. It may be described as Positional (PDOP), Geometric (GDOP), or Horizontal (HDOP) and all indicate positional accuracy. A good DOP (normally indicated on equipment by a lower number) means greater accuracy as the cut of satellites is larger, and vice versa. A number <1 is ideal and will give the highest confidence level, but in normal navigation a figure of 2-5 is rated 'Good' and this may be seen as the minimum threshold for making accurate decisions. If figures are above this, and especially above 10, then the mariner should be considering discarding the position fix and using other means to cross check as it may only be taken as a rough estimate of the vessel's position.

4.6.2 Receiver Autonomous Integrity Monitor (RAIM)

RAIM is a methodology built into receivers to give indication on GNSS fix integrity to the user, sometimes using a red, amber, green traffic light system. (Note, as previously mentioned some older DGPS, and more modern SBAS receivers may use a red or green system to indicate if corrections are being received, and if so, caution should be shown to what each light is showing). The basic principle works much in the same way as traditional visual fixing, when interrogating the quality of a fix, or 'cocked hat'. RAIM uses the solution residuals; these are the differences between the distance to each GPS satellite compared to the range the receiver has measured. If these residuals are small and stay within a set threshold, then the system will indicate green; and if greater than the threshold the red alarm will be displayed. RAIM normally requires at least 5 satellites per fix to correctly determine the fix accuracy and any position lines that are anomalies.

In cases of poor satellite geometry (also indicated by a higher DOP figure) the amber warning should display to show that, although the fix may be accurate, it is diluted based on the cut of the position lines. This is to alert the mariner that RAIM will not function correctly with poor geometry.

The key element is the size of the detection threshold, which is often set automatically, and is something the mariner has no control over. It is important that this is correctly balanced, as if it is too small even good fixes will be shown as an error, and too large will allow poor fixes to go unnoticed. Receiver standardisation and testing is important to ensure the receiver's RAIM algorithm sets the alarm condition correctly.

4.7 Datums

GNSS positions are referenced to the World Geodetic System 1984 Datum (WGS 84). This may not be the same as the horizontal datum of the chart in use, meaning that the position when plotted may be in error. The receiver may convert the position to other datum; however, these facilities should be used with caution. In this case the observers must ensure that they are aware of the datum of the displayed position. Where the difference in datums is known, a note on the chart provides the offset to apply to positions referenced to WGS 84 for plotting on the chart, but where this offset is not provided, the accuracy of the plotted position should be treated with caution.

Mariners must read the note on satellite-derived positions on the Admiralty charts for more information. Further information can also be found in the **Mariner's Handbook (NP 100)** and in **Annual Summary of Admiralty Notices to Mariners, No19**.

5. Electronic Navigation Systems and Charting

5.1 General

An Electronic Charting System (ECS) in its broadest form includes any electronic system that can be used for charting in the marine environment. Those that comply with the IMO requirements for SOLAS class vessels are known as the Electronic Chart Display and Information System (ECDIS), and all other types of electronic chart systems, retain the general classification of ECS. If an ECS is carried on board, the continuous use of up-to-date paper charts remains both the primary method for safe navigation and to fulfil the SOLAS chart carriage requirements. It should be noted that a vessel required under SOLAS Chapter V to carry an ECDIS system may still carry paper charts as a backup. The only exception to this will be if the vessel chooses to equip itself for full paperless navigation, which will see this backup arrangement be fulfilled by another ECDIS system.

If ECDIS is used to satisfy the chart carriage requirements of SOLAS Chapter V, ECDIS must, where available, use Electronic Navigational Charts (ENCs). These are vector charts produced to International Hydrographic Organization standards and are officially issued by (or on the authority of) a government authorised Hydrographic Office or other relevant government institution. There is full ENC coverage for UK waters and the vast majority of the rest of the world now has ENC coverage, but some anomalies may exist and careful checking during the planning process should be conducted in consultation with relevant national Hydrographic Offices (HOs) as necessary. Coverage overview can be checked on the IHO website, but navigating officers should consult their regional HOs or chart suppliers during planning.

<https://iho.int/en/iho-online-catalogues> (<https://iho.int/en/iho-online-catalogues>)

In rare cases where ENC data is not available; Raster Navigational Charts (RNC) may be used with the ECDIS in the Raster Chart Display System (RCDS) mode. However, when operating in RCDS mode, the RCDS must be used in conjunction with an appropriate folio of up-to-date paper charts. **MGN 285 (as amended)** provides further guidance.

Further comprehensive guidance on the use of ECDIS is given in the IMO publication, **MSC.1/Circ.1503/Rev.2: ECDIS – Guidance for Good Practice**.

An ECS that does not meet the specific performance and hardware standards of ECDIS will not be approved for primary navigation, and any ECS (including an ECDIS) not using ENCs will not be approved for primary navigation, and as such paper Standard Navigation Charts (SNCs) are required. The focus of the text in this section leans towards ECDIS and ENC use for that reason.

Mariners on UK code vessels below 24m in length may have the option to use a 'Mini-ECDIS' system which, when used with ENCs may be used as a primary means of navigation (including paperless) if the appropriate requirements of the system, vessel, and training are met. **MGN 319 (As amended)** provides guidance.

5.2 ECDIS Sensors

ECDIS is integrated with the GNSS, one of the required three inputs along with gyro and speed/distance log, which continuously and automatically feed data, enabling the vessel's position to be continuously displayed.

Care shall be taken by the mariner to ensure the data feeds being inputted into the ECDIS are accurate and valid.

5.3 Electronic Navigational Charts (ENCs)

An ENC is a vector chart, comprising a database of individual items of digitised chart data which can be displayed as a seamless chart. They contain all the chart information necessary for safe navigation and ENCs of appropriate detail are provided for different navigational purposes such as coastal navigation, harbour approach and berthing. The amount of detail displayed is automatically reduced when the scale of a particular ENC is reduced, to lessen clutter. Individual items of data can be selected and interrogated, and all relevant information will be displayed (for instance, all the available information relevant to a light or navigation mark).

With vector charts the data is “layered”, enabling the user to de-select certain categories of data, such as textual descriptions, which may clutter the display and may not be required at the time. It is also possible for the user to select a depth contour which provides an electronic safety contour that can automatically warn the mariner when approaching shallow water. Mariners should use the facility to de-select data with extreme caution as it is possible to accidentally remove data essential for the safe navigation of the vessel. Mariners should also be aware that the ECDIS ‘display base’ does not provide sufficient information to support safe navigation, and if this setting is selected, or displayed from startup, the display should be built to include appropriate content. Individual manufacturers may include certain display pre-sets which may be selected to show chart features.

The nature of vector charts and the layered data within them means that they are available to be used with several features within ECS to aid the mariner in navigation tasks and due to their increased functionality it can be seen that ENCs are therefore very much more than an electronic version of a paper chart. ENCs are designed to aid situational awareness and reduce burden to increase safety and are detailed later in this section.

Some HOs, including the UKHO, are starting to include extra contours in the ENCs to support the selection and use of safety contours in charting systems, especially ECDIS. These are sometimes known as High Density ENCs, or HD ENC. Overall appearance is largely unchanged, but there will be increased functionality with the safety contour feature of ECDIS. UKHO HD ENCs will not require additional or special purchase, and will be included in the chart catalogue as per any other ENC.

5.4 Raster Navigational Charts (RNCs)

The Raster Chart Display System (RCDS) uses RNCs, which are exact facsimiles of official paper charts, and for which Hydrographic Offices take the same liability as for their paper products. RNCs, being little more than images, do not have the functionality of ENC's that allow ECS such as ECDIS to interact with them and alert the operator. Therefore, the availability of safety features in the charts themselves is nil, and if navigating on them the mariner should treat them just as they would a paper SNC. RCDS will still allow the constant position of the ship to be plotted on the chart from sensors, and some systems may allow manual corrections to be drawn/added, but extreme caution must be given in this mode. As previously mentioned, in these cases when ENC's are not in use, an appropriate and up-to-date folio of paper charts is required as the primary navigation means.

RNC coverage may be available in some areas of the world where ENC coverage is not, and the mariner should refer to **MGN 285** for guidance of operation in these cases. The allowance of **MGN 285** is not applicable in UK waters, which has full ENC coverage. Further detail is given in Annex 2 of **MSC.1/Circ.1503/Rev.2: ECDIS – Guidance for Good Practice**.

5.5 Chart maintenance and catalogues

The principles of ordering and maintaining a folio of charts is the same for paper SNCs and for ENC's; sufficient charts should be carried for the area being navigated and should be kept up to date. **MGN 610** provides details on the SOLAS requirements. ENC's are displayed as 'cells' of different scales, like individual paper charts, but they may not follow the same boundaries or cover the same area as corresponding paper SNCs, even from the same HO (Hydrographic Office). They are purchased on an individual basis, with a 'licence' for a set period (normally in months). This licence will allow the chart to be installed, loaded, and displayed.

However, a critical feature which the Mariner should be aware of is that when the license expires the chart may still be displayed if not removed from the system but will not receive updates and therefore will not be valid for navigation. The mariner should be aware of the state of their licences and the checking of such should form part of the passage plan appraisal process.

Non-ECDIS systems, or any system using non-official electronic charts, will see differences in how charts are purchased (maybe in geographical area rather than individual cells), updated, and loaded into the system. The mariner should be aware of the specifics of the folio they are using and the coverage provided by the charts they have purchased.

It would be prudent for a mariner to order charts immediately bordering any operating area to allow for contingency if the vessel deviates from the plan, but it should also be noted that small harbours or ports which may have been displayed as an 'inset' on a paper chart will likely be a separate ENC cell. It may not be required or appropriate for certain vessels to hold these largest scale charts when navigating on passage and it could congest the ECDIS system. Therefore, the mariner should check the order and cell holdings, especially if using chart software that automatically generates an order based on an imported route.

Updates to official ENCs will be provided by the issuing HO, and there are various ways of receiving and transferring this into the ECDIS system depending on ship connectivity and equipment manufacturers. It is recognised that updates may not always be possible on a strictly weekly basis due to operational reasons, but these periods should be minimised where possible and mariners must be aware of the update status of the charts they are using. This does not relieve any responsibility of the mariner of monitoring navigational warnings, e.g., NAVTEX, and ensuring they are properly documented and appended to charts. Updates are provided cumulatively, so only the most recent weekly update needs to be applied, which may be useful after a docking period or a period when the vessel is unused. When new charts are purchased, they should be the latest editions, but it is prudent to apply the latest weekly update after chart install to ensure safety.

5.6 ECDIS Performance Standards

An operational ECS comprises of hardware, software, and data, and for an ECDIS all of these must meet international performance standards. It is important for the safety of navigation that the application software within the ECDIS works fully in accordance with these standards and can display all the relevant digital information contained within the ENC. Although it is not necessarily the responsibility of the mariner, as the user, to ensure their equipment conforms to the performance standard at a technical level, they should be aware if their equipment is classed as compliant and if official and updated ENCs are in use. They should also be aware of the possible consequences of using un-official hardware or software in relation to compliance.

SOLAS regulation V/16 requires that there are adequate arrangements in place to ensure that the performance of navigational equipment required by SOLAS Chapter V is maintained. Any noted deficiencies and evidence of the record of maintenance of the defective equipment should be readily available. The IMO circular **Guidance on Procedures for Updating Shipborne Navigation and Communication Equipment (MSC.1/Circ.1389)** refers to general navigational equipment, and specific EDCIS maintenance guidance is now included at Part B of the IMO circular **MSC.1/Circ.1503/Rev.2: ECDIS - Guidance for Good Practice**.

The latest versions of relevant IHO standards are available at the weblink below. An ECDIS which is not upgraded to be compatible with the latest version of the performance standards or the S-52 Presentation Library may be unable to correctly display the latest charted features. Additionally, the appropriate alarms and indications may not be activated even though the features have been included in the ENC.
<https://iho.int/en/standards-in-force> (<https://iho.int/en/standards-in-force>)

5.7 ECDIS Alarms and Indicators

As part of prudent navigation using ECDIS, it is important to be adequately familiar with the types of alarms available within the system.

The definitions of the Alert, Alarm and Indication are described below, and the user should be aware of the differences, and which events will activate which alert type as listed within the ECDIS performance standard.

(i) Alert - Alerts announce abnormal situations and conditions requiring attention. Alerts are divided in four priorities: emergency alarms, alarms, warnings, and cautions.

(ii) Alarm - An alarm is a high priority alert. A condition requiring immediate attention and action, to maintain the safe navigation and operation of the ship. An alarm will be both audible and visual and will only cease when acknowledged by the mariner.

(iii) Indication - Visual indication giving information about the condition of a system or equipment. Some systems may also have a brief, self-cancelling, audible alert as part of an indication.

5.7.1 Alarm fatigue and excess alarms

With the increasing digitalisation of navigation bridges, it is inevitable that excess alarms will be encountered. It is important that equipment is set up correctly, taking into account external factors such as traffic density,

proximity to danger, and environmental conditions to try and minimise this. Mariners should be intrinsically aware of the ECDIS alerts and how to differentiate them, but also the alarms of other bridge systems and the fact that they may all sound similar, that multiple may alarm for the same event, and that many will be integrated into the bridge alert management systems (BAM).

Alarm fatigue is an extremely hazardous consequence of excess alarms where a watchkeeper may be silencing alarms without checking them (a crying wolf scenario) or may become either complacent or distracted, resulting in a critical notification being missed. The mariner should be acutely aware of the hazards and any mitigations they can put in place when managing their navigational watch.

5.8 Overreliance on GNSS for position fixing.

The mariner should be aware of the guidance stipulated within section 4.5 of this document and shall ensure that independent means of position fixing is utilised whilst navigating using ECDIS systems. Section 5.10 also states how radar overlay onto the ECDIS system and an ENC can help visually verify the position accuracy of the GNSS input into ECDIS.

5.9 Navigation planning and execution

The Mariner shall have due regard to the specific elements of electronic navigation which differ from traditional paper navigation when planning a route using an ECS. The following section gives an overview of some of the most common and important safety aspects of electronic charts but is not exhaustive and it is recommended that detailed, company and vessel specific guidance is made available within a vessels Safety Management System.

5.9.1 Safety contour, soundings, shades

The safety contour feature is a key element of electronic navigation and is a mandatory feature within ENCs and ECDIS. Other forms of ECS may have similar features but may not comply with the same performance or display standards of ENCs and ECDIS. When the mariner sets the safety contour settings, the system will highlight the corresponding contour on the chart, or if not an exact match, the next deepest contour. This will display visually in bold, and all areas shallower will be classed as danger and will display in different colour shades to aid visual identification.

ECIDS will interact with this area and contour, alerting the mariner if they approach or cross it. Safety depth can also be set, which will highlight all soundings shallower than the exact setting. Some ECDIS systems may combine this into one setting, but in any case, the navigator shall calculate the safety depth for their vessel, including safety margins, and set the settings exactly so the danger areas can be displayed.

HD ENC's may mean that a more precise contour can be selected by the system to allow more accurate display of safe water, but until full coverage is available caution should be exercised as when changing to a new chart the safety contour may jump to a deeper contour.

5.9.2 Category zones of confidence (CATZOC)

Source data diagrams on paper SNCs have been replaced on electronic charts with other methods of displaying the type and accuracy of survey data. With ENC's this is CATZOC which are visual graphics displaying the survey quality of areas on the chart. CATZOC values are displayed using triangular or lozenge-shaped symbol patterns containing stars. The number of stars relates to the level of data quality with six stars equalling A1, being the most accurate, and stars equalling D, the lowest data quality. A single star is not used to avoid possible confusion with a rock symbol. If an area has not been assessed for CATZOC a U symbol is used with equals unassessed. **The Mariner's Handbook (NP100)** provides detailed information on the characteristics of all CATZOC levels.

Despite vastly increased global ENC coverage, the survey standards of the data contained within many of them will remain below the level of modern standards due to the age of the soundings. When operating in an area of CATZOC B or worse, especially in shallow waters, it may be prudent to allow additional safety margins in planning and adding to the safety settings mentioned in 5.7.1. Guidance on the depth accuracy and additional safety margin required with the varying CATZOC levels can be found within **The Mariners Handbook (NP100)** as well the **IHO Mariners Guide to Accuracy of Depth Information in Electronic Navigational Charts (ENC)**.

5.9.3 Cross track distance (XTD)

Another feature to assist the watchkeeper within ECDIS is route monitoring using constant position plotting. The XTD will display in numerical form on many, if not all, of the display presets to give quick indication to the watchkeeper of their lateral separation from the navigation route. Settings can be made to allow ECDIS to indicate or alarm when certain separation thresholds are reached.

The mariner shall ensure that the XTD does not exceed the agreed Cross Track Corridor (XTC) described below.

5.9.4 Cross Track Corridor (XTC)

The mariner should set a cross track corridor (XTC) along each route leg to allow a margin for the vessel to safely navigate the vessel within. This may be adapted to different sizes on individual legs to correlate with the area being navigated.

When a route is created, and the XTC set, the navigator should utilise the scan/validation function within the ECDIS software to check the route. This function will detect any hazards within the XTC below the safety depth and alert the mariner. Any Hazards identified by the software during the scan/validation function must be accepted by the operator before continuing. There are some important safety considerations to be made:

(i) The safety depth must be set correctly for the vessel and voyage before this check is conducted.

(ii) The system **will not** scan outside the XTC, therefore hazards outside this distance **will not** be detected or alarmed. This means, should the vessel stray outside the XTC it may not be in safe waters. Sufficient safety distance from the edge of the XTC to the safety contour should be allowed, and it may be prudent to set a wider XTC when scanning/verifying the route and then reducing this to normal levels for the watchkeeper to navigate within. This will allow some safety margin if the XTC needs to be crossed before a re-scan can take place.

(iii) The XTC used during passage monitoring should be assessed against the prevailing hazards of the passage leg and the width of the corridor amended accordingly ensuring sufficient redundancy within the corridor to allow for route deviation caused by collision avoidance or other prevailing conditions.

(iv) The Mariner should be aware that any deviation from the XTC will place the vessel into an area that has not been validated. Subsequently, this means that the area has not been assessed as safe and therefore the passage plan should be re-appraised.

(v) The scan/verify function will pick up all dangers within the XTC (provided they meet the safety depth criteria) regardless of if they are displayed physically or not. This means in base display mode, safety is not compromised but does not relieve the mariner of the requirement to ensure feature display settings are correct as per 5.3 of this notice.

It is recommended that clear guidance on use of cross track corridors is made available within the vessel's Safety Management System.

5.9.5 Anti-grounding and look-ahead features

These features may vary in name and style between individual charting systems but will allow the mariner to set limits that the system will check as the vessel's position moves and alert if dangerous objects are detected, giving early warning of hazards. In most cases there will be a fixed 'box' around the vessel and, once dimensions are set, will provide the last layer of notification, and additionally watchkeepers may use look-ahead functions to scan further ahead. These will normally be specified by a time and angle rather than distance, giving a cone ahead of the vessel in whichever direction the SOG is progressing. Being time based, this will vary proportionally to the SOG so as to provide a much greater zone of detection at higher speeds and maintain a suitable warning time throughout.

It is recommended that all bridge officers are familiar with the setup and operation of the features specific to the ECS on their bridge, and appropriate default or minimum settings are available in the vessel's Safety Management System. It may be appropriate to increase the anti-grounding settings if the watchkeeper deems it necessary and they should not be afraid to do so and ask for guidance if needed.

5.9.6 Chart display and features

As previously described in 5.3 of this section electronic charts, especially ENC's contain layers of data and although they provide all the same information as paper SNC's, they sometimes display this in different ways, and often have additional features or information. Symbology, such as buoyage, by default is shown in a traditional manner but this can be changed to simplified forms for ease of electronic viewing. There are also various charting symbols that are unique to ENC's or differ from SNC's. The watchkeeper should be aware of these by referring to **Admiralty Guide to ENC Symbols used in ECDIS (NP5012)**.

The pick report or query function allows the mariner to see data such as light characteristics, names, and elevations that are hidden in ENC at times to avoid clutter. Chart notes which would normally be printed in dead space on an SNC are contained as separate text files with ENC's and the mariner should familiarise themselves with searching and using this information. Various other settings will be available (for example extended or full-length light sectors) but as previously mentioned, the display can quickly become cluttered, and care should be taken when they are toggled on or off. Due to the vast amount of data held, even displaying full soundings may be cluttering, especially when zooming out from the default scale and dangers in the vicinity may not be obviously distinguishable.

5.9.7 Colour schemes

ECS will have different colour schemes available for selection, and in ECDIS these will be strictly prescribed in conjunction with ENC's to ensure

proper viewing in day and night conditions. Non-official ECS may not conform with full colour palettes and care should be taken. The correct night scheme should be in use during hours of darkness to aid with night visibility, but if left on in daytime can mean that safety critical information is not obviously distinguishable. Display monitors may have independent 'dimming' capability, but this should be used with particular caution and if used then full brightness is to be restored during daylight hours. These principles will be the same for many other bridge equipment, especially radar.

5.9.8 Conclusion

Proper use of features such as those described above, especially the safety contour and look-ahead, are key to ensuring potential benefits of ECDIS are being utilised by the mariner, and therefore by reducing burden and improving overall navigational safety. The Mariner shall ensure these settings are regularly checked and adjusted to suit the environment to ensure continued optimisation of these safety features. It is also recommended that Bridge Navigational Watch handover procedures and checklists include ECDIS setup and settings check.

It is recommended that when not directly using an ECS the display should be left centred on the vessel position, at the default (1:1) scale for the chart in use to aid easy interpretation by the mariner when checking the display.

5.10 ECDIS Overlays

Many ECDIS or other ECS systems have the ability to overlay layers of information from other bridge equipment. Current common overlays are Radar (Target Tracking only or full graphical overlay) and AIS, although these are likely to increase in the future as new technology enters use. This can greatly aid the mariner in building a situational picture, however caution should be taken that this does not obscure essential information from the chart, especially when using radar full graphical overlay.

Radar overlays can provide methods of verifying the position input to the ECDIS/ECS in coastal navigation, either using the graphical overlay or transferring known fixed points acquired as targets on the radar.

When using electronically overlaid data from ARPA, ATA, or EPA, or when navigating with part or all the radar display overlaid or underlaid on the chart display, there is a danger that the combined display may become over cluttered with data. The overlay of target data on an electronic chart does not reduce the need for the targets to be observed on the radar

display. Although radar targets may be interrogated on the ECS display to give details of CPA for example, the information for collision avoidance decision making should be made from that on the radar unit which will not encounter delays or errors that may be seen once data is transferred to an ECS.

Mariners should exercise caution where target vectors based on the vessel's course and speed through the water are overlaid on an electronic chart which displays the vessel information based on course and speed over ground.

5.11 System based datum conversions

Manufacturers of GNSS receivers, and ECDIS/ECS often incorporate a user selectable datum transformation capability into their software. This capability enables users to deal with datum differences in a systematic and apparently automatic manner. Whilst this might appear to be a good thing, considerable caution needs to be exercised.

A potential problem is that interoperability issues might emerge when connecting a GNSS receiver to an ECDIS or other ECS, particularly if the receiver is configured to convert its position output to a local or regional datum. Care must be taken to ensure that GNSS receivers are configured to provide position in the datum that is expected by the ECDIS. In most cases this will be the WGS84 datum, but manufacturer's instructions should always be carefully consulted to ensure correct system operation.

Further, if using an ECDIS, it is conceivable that a datum correction might be applied twice; once by the GPS receiver and again by the ECDIS. Once again, system manufacturer's instructions should be consulted to ensure this problem is avoided.

5.12 Backup

ECDIS with adequate backup arrangements may be accepted as complying with the requirements of SOLAS V/19 and 27. Accepted backup arrangements are either a full paper chart folio or separate ECDIS system, utilising a separate power supply and designed to ensure safe navigation is not compromised in the event of the failure of the main ECDIS system. Further details on the requirements of a backup ECDIS system can be found in **IMO Resolution MSC.530(106)**.

Mariners should be aware of the backup systems on their vessel relating to ECDIS, including the operation reversionary sensors and how their input to ECDIS may differ from the primary source. Checking and testing of secondary sources of navigational data into ECDIS should be as regular as the primary sources to ensure minimal disruption if they are required.

If multiple ECDIS systems are carried they must be powered from power sources that are independent of one another, and all will be linked to emergency power generation. This may include uninterruptable power supplies (UPS) to allow seamless transition to emergency systems, and therefore minimal reduction in navigation capability should an incident occur.

Regardless of backup arrangement in use, whilst monitoring an active route, the Mariner shall ensure that this arrangement is utilised to the same extent as the primary navigation source.

6. Automatic Identification Systems (AIS)

Perhaps the most prolific emergence of an electronic navigational aid over recent years has been the introduction of AIS. AIS was introduced as a navigation aid for situational awareness, with a focus on shore-based monitoring also, such as VTS. **IMO Resolution A.1106(29)** provides the clarity on this matter, specifically in parts 41.1 and 42.2.

However, it is the UK policy that AIS is explicitly not to be relied upon solely for collision avoidance and does not relieve the mariner of their responsibility under COLREG when determining if risk of collision is developing or exists. **MGN 324 (As amended)** provides the guidance on this. Mariners on UK vessels, and any vessel within UK waters should be aware of this policy, and the hazards and limitations of making decisions based solely on AIS data.

SOLAS chapter V gives details on mandatory carriage, with UK guidance in **MGN 610**, but the rapid reduction in production costs and development of micro technologies means that AIS systems are widely available to the public, thus a vast number of vessels now operate with it. However, mariners should be aware that not all vessels are required to be fitted with AIS and, therefore, not all transmit AIS data. In addition, it is possible that not all the AIS data displayed will be accurate, particularly data which is inputted manually on the target vessels' AIS units.

Class A transmitters are required for mandatory carriage and provide far greater information about the vessel than Class B transmitters which are more likely to be fitted to recreational or fishing craft. Class B transmitters often are 'transmit only', and do not constantly transmit like Class A, often only showing for a few minutes at a time meaning data can be considerably out of date.

Mariners on vessels fitted with AIS should ensure it is updated and properly transmitting at all times when not made fast to the shore, with specific reference to the "navigation status" within the voyage data section. The data transmitted is only as accurate as what is entered, and errors are frequently observed.

Caution should be placed upon data received from AIS linked Pilot Plugs connected to PPU units. Pilot Plug data is not controlled and loosely regulated. Additionally, users should be aware that the antenna position can be manually offset from the actual location, which can lead to adverse predictions with vessel track projections. Other common errors to consider include speed and position smoothing.

The VHF data exchange system (VDES) is in the early stages of development and is a likely successor to the AIS system. It is expected to maintain almost all the features of AIS but introduce additional data exchange features and allow greater bandwidth to reduce strain on the system capacity as seen with AIS at present. Further updates to VDES and requirements for carriage will be promulgated when appropriate.

7. Conclusion

The accuracy and functionality of electronic navigational aids has increased considerably in recent years and with international goals and ambitions for full digitisation, this is unlikely to change in the future. However, there is still a danger that over reliance on the output from a single item of equipment may lead to an accident. The need to cross check the vessel's position using other means is as important today as it ever was, as is the basic requirement under Rule 5 of the COLREG to maintain a proper lookout. Accidents have occurred with ships equipped with the best equipment but where mariners have been over reliant on the equipment output, and disaster could have been averted by the simple expedient of maintaining a proper lookout, knowledge of the equipment and how solutions are generated, and cross-checking of the solutions provided by the electronic aids.

8. Relevant MCA references

- MGN 285 (M+F) (<https://www.gov.uk/government/publications/mgn-285-electronic-charts-operating-ecdis-in-the-rcds-mode>) - Electronic Charts: The Use of Risk Assessment Methodology when Operating ECDIS in the RASTER Chart Display System
- MGN 293 (M+F) (<https://www.gov.uk/government/publications/mgn-293-alternative-arrangements-for-paper-chart-requirements>) - Alternative Arrangements for Meeting Paper Chart Carriage Requirements on MCA Code Vessels under 24 metres in Length and Fishing Vessels under 24 metres in Length
- MGN 313 (F) (<https://www.gov.uk/government/publications/mgn-313-keeping-a-safe-navigational-watch-on-fishing-vessels>) - Keeping a Safe Navigational Watch on Fishing Vessels
- MGN 315 (M) (<https://www.gov.uk/government/publications/mgn-315-keeping-a-safe-navigational-watch-on-merchant-vessels>) - Keeping a Safe Navigational Watch on Merchant Vessels
- MGN 319 (M+F) (<https://www.gov.uk/government/publications/mgn-319-mf-amendment-1-electronic-chart-plotting-systems-for-small-vessels>) (As amended) - Acceptance of Electronic Chart Plotting Systems for Fishing Vessels under 24 Metres and Small Vessels in Commercial Use (Code Boats) up to 24 Metres Load Line Length
- MGN 324 (M+F) (<https://www.gov.uk/government/publications/mgn-324-mf-amendment-2-navigation-watchkeeping-safety-use-of-very-high-frequency-vhf-radio-and-automatic-identification-system-ais>) (As amended) - Navigation: Watchkeeping Safety - Use of VHF Radio and AIS
- MGN 360 (M+F) (<https://www.gov.uk/government/publications/mgn-360-changes-to-routeing-measures-in-electronic-navigational-charts-encs>) - Navigation: Implementation of Changes to Routeing Measures in Electronic Navigational Charts (ENCs)
- MGN 364 (M+F) (<https://www.gov.uk/government/publications/mgn-364-mf-amendment-1-navigation-traffic-separation-schemes-application-of-rule-10-and-navigation-in-the-dover-strait>) (As amended) - Navigation: Traffic Separation Schemes - Application of Rule 10 and Navigation in the Dover Strait
- MGN 369 (M+F) (<https://www.gov.uk/government/publications/mgn-369-navigation-in-restricted-visibility>) - Navigation: Navigation in Restricted Visibility

- MGN 610 (M+F) (<https://www.gov.uk/government/publications/mgn-610-mf-solas-chapter-v-guidance-on-the-merchant-shipping-safety-of-navigation-regulations-2020>) - Navigation: SOLAS Chapter V - Guidance on the Merchant Shipping (Safety of Navigation) Regulations 2020
- MSN 1781 (M+F) (<https://www.gov.uk/government/publications/msn-1781-amendment-1-the-merchant-shipping-regulations-1996-amendments-to-annex-iv>) (As amended) - The Merchant Shipping (Distress Signals and Prevention of Collisions) Regulations 1996: COLREG

Note: the above are correct at the time of publishing, subsequent amendments or new editions may be in force at time of reading.

More information

Technical Services Navigation
Maritime and Coastguard Agency
Bay 2/24
Spring Place
105 Commercial Road
Southampton
SO15 1EG

Email: navigationsafety@mcga.gov.uk

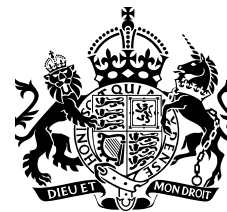
Website: www.gov.uk/mca (<https://www.gov.uk/mca>)

Please note that all addresses and telephone numbers are correct at time of publishing.



OGI

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